

Day 07 - Linux Fundamentals Revision

Objective

To reinforce the Linux concepts learned during the past three days by revising key topics, practicing essential commands, and strengthening the foundation required for Cloud Computing and DevOps.

Why Revision is Important

Learning Linux is not just about memorizing commands—it is about understanding how the operating system works and becoming comfortable using the terminal.

Regular revision helps to:

- Improve command-line confidence
 - Remember frequently used commands
 - Strengthen problem-solving skills
 - Build a solid foundation for advanced topics
 - Reduce mistakes while managing Linux servers
-

Topics Revised

Linux Basics

Revised the core concepts of Linux, including:

- What is Linux?
- Features of Linux
- Why Linux is widely used
- Open Source philosophy

- Linux distributions

Linux File System

Reviewed the hierarchical file system structure.

Important directories:

Directory	Purpose
/	Root Directory
/home	User files
/etc	Configuration files
/bin	Essential commands
/usr	User applications
/var	Logs and variable data
/tmp	Temporary files
/dev	Device files

Linux Architecture

Revised the Linux architecture layers.

```
+-----+
| User Applications |
+-----+
| Shell            |
+-----+
| System Utilities |
+-----+
| Linux Kernel     |
+-----+
| Hardware         |
+-----+
```

Responsibilities

- Hardware executes operations.

- Kernel manages system resources.
 - Shell interprets user commands.
 - User Applications provide software functionality.
-

Linux vs Unix

Linux

- Open Source
- Free to use
- Community developed
- Widely used in Cloud Computing

Unix

- Mostly Proprietary
 - Commercial licensing
 - Enterprise operating systems
 - Vendor maintained
-

Process States

Reviewed the lifecycle of Linux processes.

State	Description
Running (R)	Currently executing
Sleeping (S)	Waiting for an event
Waiting (D)	Waiting for I/O operations
Stopped (T)	Execution paused
Zombie (Z)	Process finished but awaiting cleanup

Linux Commands Practiced

File & Directory Commands

```
pwd  
ls  
cd  
mkdir  
rmdir  
touch  
cp  
mv  
rm  
cat
```

Searching Commands

```
find  
grep
```

Process Commands

```
ps  
top
```

Permission Commands

```
chmod  
chown
```

Storage Commands

```
df -h  
du -sh
```

Utility Commands

```
history  
head  
tail  
clear  
whoami  
uname -a  
ping  
wget  
curl
```

Hands-On Practice

Today I:

- Revised all Linux notes
- Practiced frequently used commands
- Navigated the Linux directory structure
- Reviewed Linux architecture
- Studied process management concepts
- Revisited file permissions
- Strengthened command-line navigation skills

Week 1 Learning Summary

Git & GitHub

- Version Control
 - Git Workflow
 - Branching
 - Merging
 - GitHub Repository Management
-

Internet Fundamentals

- Internet Architecture
 - Client-Server Model
 - Web Servers
 - Application Servers
 - Types of Applications
-

Linux Fundamentals

- Linux Basics
 - Linux Commands
 - Linux Architecture
 - Linux File System
 - Package Management
 - Process Management
 - File Permissions
 - Advanced Linux Commands
-

Key Takeaways

- Linux forms the backbone of modern cloud infrastructure.

- Strong command-line skills are essential for Cloud and DevOps Engineers.
 - Revision is an important part of long-term learning.
 - Building a solid foundation now will make advanced topics like Docker, Kubernetes, AWS, and CI/CD much easier to understand.
-

Interview Questions

1. Why is Linux preferred in Cloud Computing?
 2. What are the layers of Linux Architecture?
 3. Explain the Linux file system hierarchy.
 4. What is the difference between Linux and Unix?
 5. What are the different process states in Linux?
 6. What is the difference between `df` and `du`?
 7. How does `grep` differ from `find`?
 8. Why are file permissions important?
-

Day 07 Summary

Today was dedicated to revising the Linux fundamentals covered throughout the week. Rather than learning new concepts, I focused on reinforcing my understanding through revision and hands-on practice. This review has improved my confidence with Linux and prepared me for upcoming topics such as Linux user management, shell scripting, networking, Docker, and AWS.

Status:  Day 07 Completed (Week 1 Successfully Completed)